Problem Set - 7

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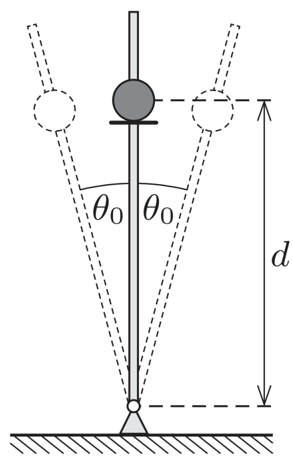
1 Mechanics

1.1 Oscillating Rod

A small smooth pearl is threaded onto a rigid, smooth, vertical rod, which is pivoted at its base. Initially, the pearl rests on a small circular disc that is concentric with the rod, and attached to it at a distance d from the rotational axis. The rod starts executing simple harmonic motion around its original position with small angular amplitude θ_0 (see figure). What frequency of oscillation is required for the pearl to leave the rod? (200 More, P21)



2 Electromagnetism

2.1 Movement in the Dipole Field

As shown in the figure, a particle of mass m and charge e moves in the field of an electric dipole $\vec{p}$, fixed at the origin of coordinates. The dipole moment is directed along the z-axis, and the dependence of the particle's position on time is given by the radius vector $\vec{r}(t)$. The angle between the radius vector of the particle and the z-axis is $\theta(t)$. (CN23, T5)

1) Let the particle perform uniform motion, in which its trajectory is a circle orthogonal to the z-axis with the center also on the z-axis. Find the speed of the particle v_0 and angle θ_0.

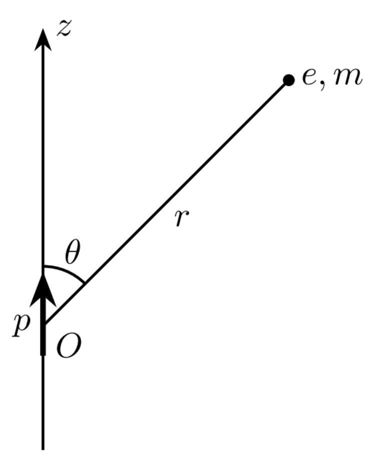
Now let the particle be stationary at the initial moment of time $t = 0$.

2) Determine the dependence of the angular momentum of a particle L on θ during its further movement.

3) Find the dependence of the radial velocity of the particle v_r on the distance to origin r during its further movement.

4) Under what condition is the initial angle θ_0 will the particle's motion be finite?

5) In the case of infinite motion, determine the dependence of distance from the origin on time $r(t)$.



3 Miscellaneous

3.1 Virial Theorem

In this problem, we will try to prove the *virial theorem*, one of the most important *statistical* theorems in physics. We are interested in the quantity

$$G = \sum_i \vec{p}_i \cdot \vec{r}_i$$

where the summation is over all particles of the system. (Kutay, reading Goldstein)

i) Find an expression for $\mathrm{d}G/\mathrm{d}t$ in terms of the total kinetic energy of the system, T, and $\sum_i \vec{F}_i \cdot \vec{r}_i$.

ii) Calculate the time average of the quantity G for some interval τ. What do we have if the motion is periodic and τ is the period of the motion? What if the motion is not periodic, but all the physical quantities of the system are finite? (Similar argument to that of the irrational period ratio in PC2024-P4.)

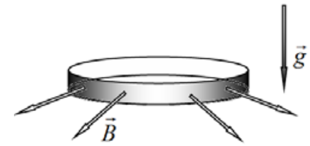
iii) The expression you have calculated in (ii) is known as the *virial of Clausius*. Use this result to derive the ideal gas law. (*Hint:* What are the assumptions for ideal gas? Where do we have $\vec{F}_i \neq 0$?)

iv) Let's investigate a case where the forces can be derived from a potential, i.e. $\vec{F}_i = \nabla V \cdot \vec{r}_i$. For a system of particles in a central force field, find an expression for the *virial theorem* in terms of T, $\vec{r}_i$, V and its derivatives.

v) What is the expression if the potential of the field is $V = \alpha r^{n+1}$? Calculate the expression for $n = -2$, i.e. for inverse-square forces.

Problem 3. Ring in a magnetic field (10.0 points)

Uniformly charged ring



A very thin ring of mass m and radius r is uniformly charged along its length with a charge q. At the initial moment of time, the ring rests horizontally and is released without a push. The subsequent motion of the ring appears in the vertical gravitational field of the Earth, characterized by the acceleration of gravity g and in the horizontal radial magnetic field of induction B. Neglect air resistance and assume that the plane of the ring remains horizontal at all times.

3.1 Find the maximum velocity of the ring center of mass v_{max} for the entire time of motion.
3.2 Find the time interval Δt elapsed from the start of the ring motion to its first reaching of the maximum velocity of the center of mass.
3.3 Find the maximum height h_{max} at which the ring center of mass falls over the entire time of motion.

Conductive ring

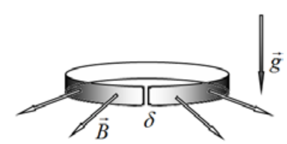
A very thin ring of mass m and radius r is made of a conductive material with a resistivity ρ and a cross section area $s \ll r^2$. At the initial time moment $t = 0$ the ring rests horizontally and is released without a push. The subsequent motion of the ring appears in the vertical gravitational field of the Earth, characterized by the acceleration of gravity g and in the horizontal radial magnetic field of induction B. Neglect air resistance and assume that the plane of the ring remains horizontal at all times.

3.4 Find the steady-state velocity v_0 of the ring center of mass after a sufficiently large period of time having passed.
3.5 The dependence of the current strength $I(t)$ in the ring on time t has the following form

$$I(t) = A_1 + B_1 \exp(\gamma_1 t).$$

Find the constants A_1, B_1 and γ_1.

Conductive ring with a cut



A very thin ring of mass m and radius r is made of a conductive material with a resistivity ρ and a cross section area s. A cut with a width $\delta \ll \sqrt{s} \ll r$ was made along the radius of the ring. At the initial time moment $t = 0$ the ring rests horizontally and is released without a push. The subsequent motion of the ring appears in the vertical gravitational field of the Earth, characterized by the acceleration of gravity g and in the horizontal radial magnetic field of induction B. Neglect air resistance and assume that the plane of the ring remains horizontal at all times.

3.6 Find the steady-state acceleration a_0 of the ring center of mass after a sufficiently large period of time having passed.
3.7 The dependence of the current strength $I(t)$ in the ring on time t has the following form

$$I(t) = A_2 + B_2 \exp(\gamma_2 t).$$

Find the constants A_2, B_2 and γ_2.

Mathematical hints for the theoretical problems

The following integrals may be useful:

$$\int \frac{dx}{ax+b} = \frac{1}{a}\ln|ax+b|.$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}, \text{ where } n \text{ is integer}$$

$$(1+x)^\gamma \approx 1 + \gamma x + \frac{\gamma(\gamma-1)}{2}x^2, \textit{ for } x \ll 1 \textit{ and any } \gamma$$

Theoretical Problem No. 1 (10 points)

Fred & Barney's car

Fred and Barney built a car having as "wheels" two identical square prisms (figure 1). The wheels are perfectly suited to the road's shape (that is a periodical repetition of identical bumps), so that the center of mass of the car does not move vertically during the trip. The wheels never slip on the road. During the car's motion, the wheel's long edge always touches a „valley" of the road; road „heights" are reached by a line passing through the half of a rectangular face of the "wheels". The vehicle's movement starts in the tops of two road bumps. Initially, the horizontal translational velocity is $\vec{v}_0$.

Figure 1

The total weight of the vehicle excluding its „wheels" is $M \cdot \vec{g}$, distributed equally on the two wheel axles. Further, assume that the only forces acting on car are gravity and normal force.

Task No. 1 - Kinetic energy of the car

1.a. Determine the expression of J - the moment of inertia of wheel reported at its own axles if its mass is m and the length of its square side is $2a$.

1.b. Determine the expression of kinetic energy of the car and the expression of the angular velocity of its wheels:

i. when the car pass on the tops of the bumps;

ii. when the car pass with each wheel through a „valley".

Task No. 2 - Road's shape

In the figure 2 is shown the motion of the lower side of the square cross-section of the wheel on a bump (the curve passing through the points x_s, T, x_d).

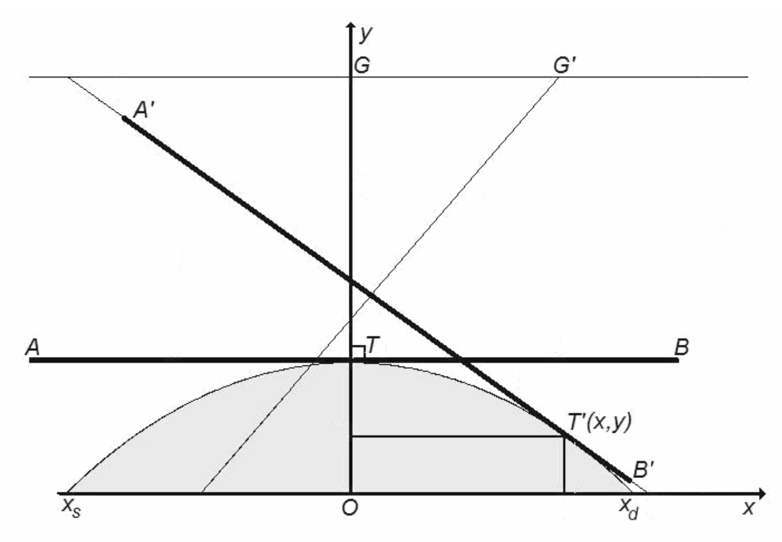


Figure 2

The road is a periodic repetition of such a bump.

The Ox axis of the coordinate system passes through valleys of the relief and the Oy axis passes through bump tip. *AB is wheels side, T is the contact point between wheel and road and G is wheel axis position – in initial position. $A'B'$, T', G' have the same meaning – but for a certain position.* In cross-section the wheel axis is always on the same vertical line with the contact point between wheel and road.

2.a. Give an explanation for this finding.

2.b. Determine the analytical form $y = y(x)$ of cross-section of the road.

Hint: The derivative $y'(x)$ of the function $y = y(x)$ is the slope of the tangent to the graph of function in x. Consider known that $\int \frac{dx}{\sqrt{x^2 - a^2}} = ln\left|x + \sqrt{x^2 - a^2}\right| + C$. We recommend to use the notation $ch(x) = \frac{e^x + e^{-x}}{2}$; $sh(x) = \frac{e^x - e^{-x}}{2}$ with property $ch^2(x) - sh^2(x) = 1$

2.c. Determine the expression of the length (on horizontal) of a bump of road.

2.d. Determine the expression of minimum distance between wheel axles Fred's car.

2.e. Decide whether a machine wheeled regular hexagonal prism conveniently chosen could run on the road whose form was determined in 2.b., without vertical displacement of the center of mass of the wheel.

Task No. 3 - Accident

Assuming that analytical expression of road's bumps is $y = k - h \cdot ch(x/a)$ where k and h are two constants.

3.a. Determine the domain of possible values of the horizontal velocity of the car, under the circumstances established in problem's statement.

3.b. Determines how the amount of heat that can be released by plastic collision of Fred's car with an obstacle depends on position of obstacle on road. Immediately after the collision the car stops.

3.c. Determine the expression of heat released by collision.

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